Aeden Grandinetti

Seattle, WA · aedeng16@uw.edu · (509) 581-7780
Portfolio: aedengrandinetti-cell.github.io — *engineering projects, research, and background (in progress)*

EDUCATION

Bachelor of Science in Materials Science and Engineering
University of Washington, Seattle — Expected June 2027
Pre-Medical track; concentration in biomaterials and bioengineering topics
Major GPA: 3.55 (in progress)

RESEARCH EXPERIENCE

Undergraduate Research Assistant | *Buddy Ratner Biomaterials Research Laboratory* — UW Seattle
May 2025 – Present

- Training to operate an SLA 3D Kloé Dilase printer; prepare PDMS and photoreactive resin vats, run sensitivity tests, and print microscopic structures for bioengineering and biomaterials research.
- Formulating novel printable biocompatible resins using photopolymerization of biopolymeric materials; troubleshoot material property challenges with iterative design.
- Design 40-micron porous lattice microstructures in Onshape (truncated octahedron, rhombic dodecahedron, gyroid); collaborate with PhD students and faculty to resolve printability and functional design issues.
- Currently learning parametric modelling to increase scaffold design capabilities for vascularization research and to account for compliance in arterial grafts (attempting parameterization of a graded gyroid).

Engineering Design Member | *Bioengineers Without Borders* — UW Seattle
October 2024 – Present

- Design and execute materials strength tests for a transtibial amputation prosthetic for pediatric patients; interpret results against international ISO standards.
- Create CAD prototypes and 3D-printed PLA testing setups for the prototyping sub team.
- Navigated funding constraints to meet all required test specifications; contributed to the team's HIC competition submission, receiving substantive feedback from industry judges.
- Support the business and project-management side of engineering in addition to technical work.

CLINICAL EXPERIENCE

Clinical Volunteer | *Virginia Mason Medical Center – Vascular Surgery* — Seattle, WA
October 2025 – Present

- Provide direct, face-to-face patient interaction in an active clinical environment; assist with patient transport and prioritize genuine human connection with patients beyond task-based care.
- Follow proper PPE and infection-control protocols, including room sanitization.
- Completed code response training

Clinical Observer – Nephrology & Transplant | *Shadowing Nephrologist* — Seattle, WA
February 2026 – Present

- Observe transplant evaluations, rounding, and direct patient care in the ICU and dialysis clinic from the provider perspective.
- Bring a uniquely informed patient lens as a kidney transplant recipient, deepening understanding of care for patients with end-stage renal disease (ESRD) and chronic kidney disease (CKD).
- Taking detailed notes on observations in the clinic for reflections later on

- Reading complex literature on topics such as kidney transplant immunology and clinical pathophysiology of renal disorders and anatomy.

WORK EXPERIENCE

Barista | *Starbucks* — Kennewick, WA
May 2023 – October 2023

- Developed experience handling food in compliance with safety regulations while managing solo-staff shifts, consistently delivering quality beverages within short wait times.

Waiter & Busser | *Bella Italia Restaurant* — Kennewick, WA
November 2021 – August 2022

- Developed customer service and hospitality skills across front-of-house roles including serving, bussing, and reservations; built adaptability by quickly learning new assignments on the fly and thriving in a fast-paced team environment.

TECHNICAL SKILLS

Wet Laboratory: SLA photopolymerization, biocompatible polymer synthesis, biopolymeric resin formulation, sterile technique, solution preparation, cell culturing in agars, micro pipetting, and standard biology and chemistry wet lab procedures.

CAD & Additive Manufacturing: Onshape (lattice and parametric modelling), SLA 3D printing (PDMA and resin), Arduino circuit design.

Programming & Computing: C, Java; proficient in Microsoft Office Suite (MS Word, PowerPoint, Excel).

Clinical: Familiarity with clinical terminology and hospital work environments; patient transport, PPE protocols, room preparation and sanitization.

LEADERSHIP & ACTIVITIES

Team Captain – Science Olympiad | *Tri-Cities Preparatory High School* — Pasco, WA
September 2021 – June 2022

- Led a STEM team, managed one-on-one coaching, team meetings, club organization, and direct conversations with faculty.
- Designed and built engineering devices including an Arduino-based thermometer and salinometer, as well as a complex launching mechanism.
- Compiled team study binders and materials, leaving resources for future teams after graduation.

Judoka | *Judo* — Seattle, WA
September 2025 – Present

- Cultivating mental focus under pressure and respectful competition, which are qualities that can be directly applied to the medical field and high stress or impact situations.

PERSONAL STATEMENT

As a kidney transplant patient who has been cared for by kind doctors and nurses, I am driven to make a meaningful impact on others' lives as both a clinician and a medical researcher. My journey into biomaterials science engineering came directly from that experience, as well as from a deep pull to the intersection of medicine and engineering. Having been given a second chance at life by my wonderful donor, I have poured my newfound days into my studies, into researching in a biomaterials lab, and into independent projects exploring the state of the art in tissue scaffolding and regenerative medicine. I look forward to contributing to the medical sciences with the perspective of someone who has lived the patient experience firsthand.